National University of Computer and Emerging Sciences

Electrical Network Analysis (EE2004)

Final Exam

Date: June 6th, 2026

Course Instructor(s)

Ms. Sara Kiran

Total Time: 3 Hours

Total Marks: 100

Total Questions: 05

Student Name | Roll No | Section | Student Signature

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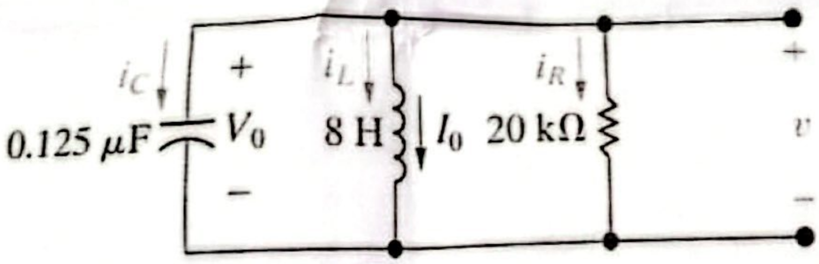
i. Attempt all the questions. Round off answers to 2 decimal places.
ii. Attempt all parts of the same question together.
iii. Show all the steps with properly labelled circuit diagrams, direct answers are not acceptable.
iv. Attempt each question only ONCE

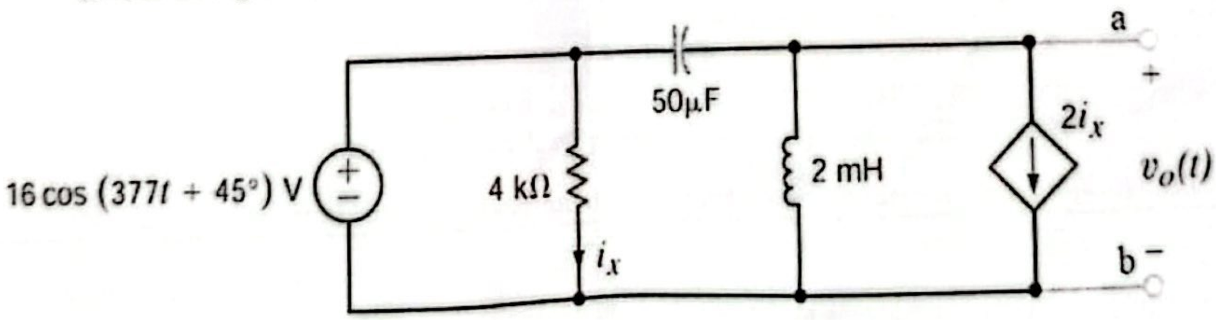
CLO #1: Apply circuit analysis techniques to determine the natural and forced response of second order circuits

Q.1 [4+16=20 marks]

a. For the parallel RLC circuit shown below, **compute** the roots of the characteristic equation and **determine** the type of response for this circuit.



b. For the Figure Shown below:



i. **Construct** the frequency-domain equivalent circuit
ii. **Apply** node voltage method to compute the Thévenin voltage with respect to the terminals a and b
iii. **Compute** the Thévenin equivalent impedance
iv. **Draw** the Thévenin equivalent circuit

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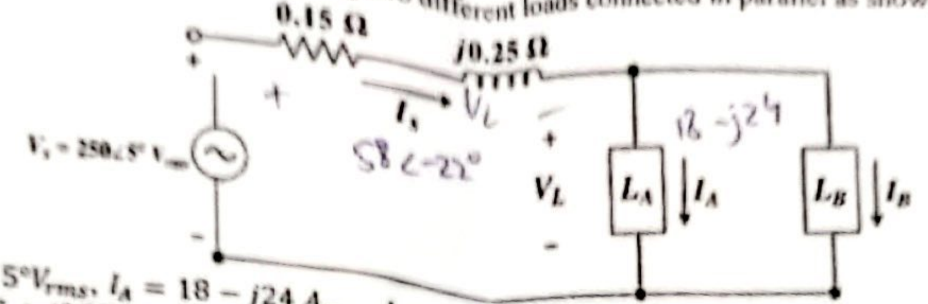
CLO # 2: Construct power triangle to compute power in AC circuits.

Q.2

An alternating power supply V_s is feeding two different loads connected in parallel as shown below. [3+6+2+3+6=20 marks]



$V_s = 250 < 5° V_{rms}$, $I_A = 18 - j24\ A_{rms}$, $I_S = 58 < -22° A_{rms}$. The transmission line impedance is $Z_{Line} = 0.15 + j0.25\Omega$.

a. **Compute** the following:
   i. I_2 *and* V_L
   ii. Complex power absorbed by each load and the complex power delivered by the source
   iii. Overall power factor of the combined load
   iv. Power loss in the transmission line

b. **Construct** power triangles for L1 and combined load

CLO # 3: Analyze balanced three-phase circuits to calculate electrical quantities.

Q.3

A balanced, three-phase circuit is characterized as follows: [4+4+4+4+4 = 20 marks]

- Y- Δ connected;
- Source voltage in the b-phase is $150 < 135°$
- Source phase sequence is abc
- Line impedance is $2 + 3j\Omega$ per phase
- Load impedance is $129 + 171j\Omega$ per phase

  a. **Construct** the single phase equivalent circuit for the a-phase
  b. **Calculate:**
     i. line currents; I_{aA}, I_{bB} and I_{cC} ?
     ii. per phase impedance of load 2
     iii. line voltages at the load end (V_{AB}, V_{BC} and V_{CA})
     iv. total complex power delivered to the Δ-connected load

CLO #4: Apply Laplace transform to determine the natural and forced response of RLC circuits.

Q4: [10+7+3 = 20 marks]

Assuming zero initial conditions in the circuit given below:

a. Transform the circuit in s-domain and **determine** the expression for voltage $V_o(s)$
b. **Compute** $v_o(t)$ for $t \geq 0$
c. **Compute** the value of $v_o(t)$ at $t = \infty$ using the final-value theorem

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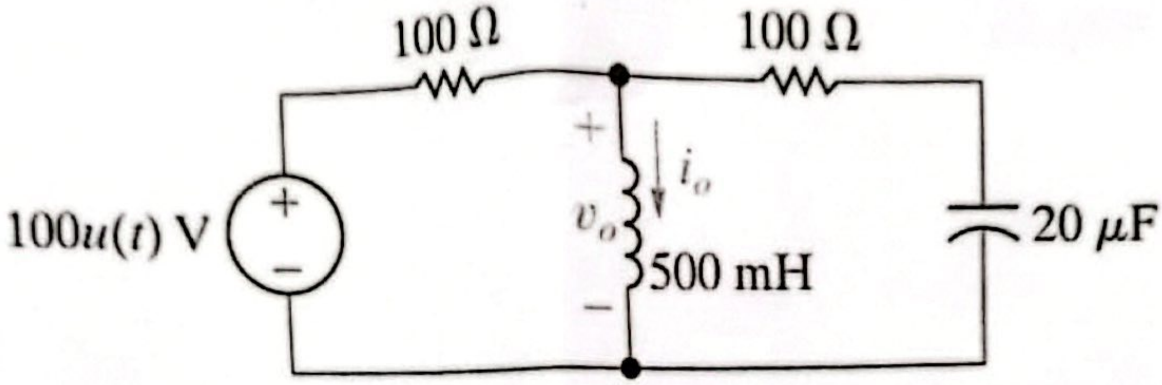
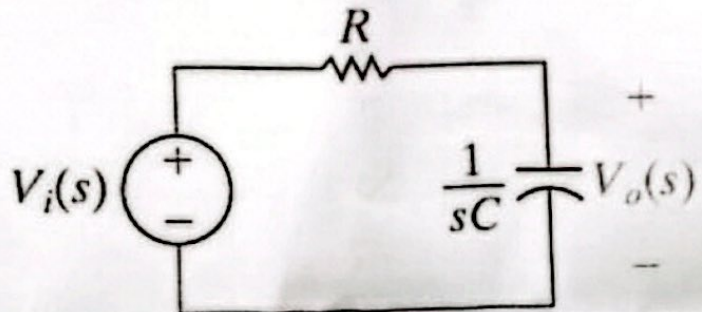


CLO #5: Analyze various types of filters using s-domain analytics.

Q5: [4+4+4+4=20 marks]

For the given filter circuit:



i. Perform qualitative analysis to **identify** the type of filter.
ii. **Determine** the expression of the magnitude and phase of the transfer function $H(s)$ for the given filter circuit and sketch the magnitude plot.
iii. **Calculate** the cutoff frequency, and the magnitude of the filter's transfer function at $\omega_c, 0.5\omega_c, and\ 2\omega_c$
iv. **Calculate** value for R that will yield a low-pass filter with a cutoff frequency of 4 kHz ($C = 1\mu F$)